Sürekli Zamanlı Fourier Transformu (Dönüşümü)

(Continuous-Time Fourier Transform)

Periyodik olmayan (aperiyodik) işaretler için Fourier dönüşümü tanımlıdır.

$$X(f) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

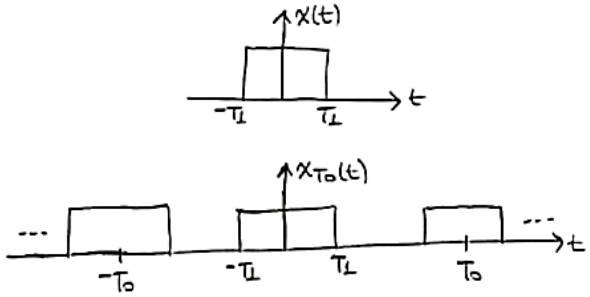
$$x(t) = \mathcal{F}^{-1}\{X(f)\} = \int_{-\infty}^{\infty} X(f)\, e^{j2\pi ft}\, df$$

$$x(t) \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\rightleftarrows}} X(f)$$

⇒ ω notasyonu: $X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega \quad , \omega = 2\pi f = 2\pi/T$$

⇒ Periyodik olmayan bir işaretin belli bir T_0 aralığında Fourier serisi açılımında T_0 sonsuza götürüldüğünde ($T_0 \to \infty$) Fourier transformu elde edilir.



$T_0 \to \infty$ iken $x_{T_0}(t) = x(t)$ dur (herhangi bir sonlu t değeri için)

$$\lim_{T_0 \to \infty} x_{T_0}(t) = x(t)$$

$$x_{T_0}(t) = \sum_{k=-\infty}^{\infty} x_k\, e^{jk\omega_0 t} \quad , \omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}$$

$$x_k = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x_{T_0}(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T_0} \int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t}\, dt$$

($x_{T_0}(t) \to x(t)$; $T_0 \to \infty$)

$$x(t) = \begin{cases} x_{T_0}(t), & |t| < T_0/2 \\ 0, & |t| > T_0/2 \end{cases}$$

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$x_k = \frac{1}{T_0} X(kf_0)$$

$$x_{T_0}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_0} X(kf_0)\, e^{j2\pi kf_0 t}$$

df=1/T0=f0

$$\Delta f = \frac{1}{T_0}$$

$$x_{T_0}(t) = \sum_{k=-\infty}^{\infty} X(k\Delta f)\, e^{j2\pi k\Delta f t}\, \Delta f$$

$$x(t) = \lim_{T_0 \to \infty} x_{T_0}(t)$$

$$= \lim_{\Delta f \to 0} \sum_{k=-\infty}^{\infty} X(k\Delta f)\, e^{j2\pi k\Delta f t}\, \Delta f$$

$X(f)e^{j2\pi ft}$'nin altında kalan alan

$$x(t) = \int_{-\infty}^{\infty} X(f)\, e^{j2\pi ft}\, df$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}$$

$$x_k = \frac{1}{T_0} X(k\omega_0)$$

$$x_{T_0}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_0} X(k\omega_0)\, e^{jk\omega_0 t}$$

$$\Delta\omega = \frac{2\pi}{T_0}$$

$$x_{T_0}(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(k\Delta\omega)\, e^{jk\Delta\omega t}\, \Delta\omega$$

$$x(t) = \lim_{\Delta\omega \to 0} \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(k\Delta\omega)\, e^{jk\Delta\omega t}\, \Delta\omega$$

$X(\omega)e^{j\omega t}$

$$x(t) = \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

⇒ Fourier dönüşümünün mevcut olma koşulları (Dirichlet condition)

1) Aperyodik x(t) işareti mutlak integrallenebilir olmalıdır, $\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$

2) Herhangi bir sonlu t aralığında,

- sonlu sayıda maksimum ve minimum değere sahip olmalıdır.
- sonlu sayıda süreksizliğe ve bu noktalarda sonlu değere sahip olmalıdır.

⇒ Frekans spektrumları:

$X(f) = |X(f)|\, e^{j\theta_x(f)}$ → $|X(f)|$: genlik spektrumu

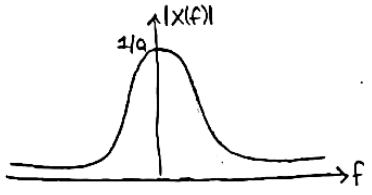
↳ $\theta_x(f)$: faz spektrumu

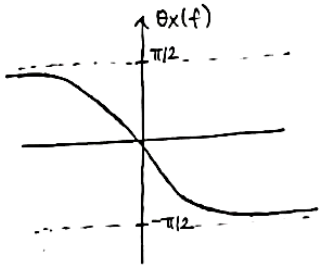
Ex: $x(t) = e^{-at}u(t)$, $a>0$ ise $X(f) = ?$ Frekans spektrumlarını çiziniz.

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt = \int_{0}^{\infty} e^{-at} \cdot e^{-j2\pi ft}\, dt = \int_{0}^{\infty} e^{-(a+j2\pi f)t}\, dt = \frac{e^{-(a+j2\pi f)t}}{-(a+j2\pi f)} \Big|_0^{\infty} = \frac{1}{a+j2\pi f} = \frac{1}{a+j\omega}$$

$$|X(f)| = \frac{1}{\sqrt{a^2 + (2\pi f)^2}}$$

$$\theta_x(f) = -\tan^{-1}\left(2\pi f / a\right)$$





Ex: $x(t) = e^{-a|t|}$, $a>0$ ise $X(f) = ?$

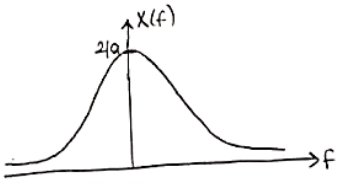
$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\,dt = \int_{-\infty}^{0} e^{at} e^{-j2\pi ft}\,dt + \int_{0}^{\infty} e^{-at} e^{-j2\pi ft}\,dt$$

$$x(t) = \begin{cases} e^{at}, & t<0 \\ e^{-at}, & t>0 \end{cases}$$

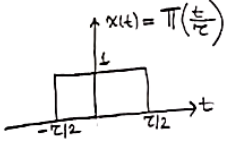
$$= \frac{e^{(a-j2\pi f)t}}{(a-j2\pi f)}\Bigg|_{-\infty}^{0} + \frac{e^{-(a+j2\pi f)t}}{-(a+j2\pi f)}\Bigg|_{0}^{\infty} = \frac{1}{a-j2\pi f} + \frac{1}{a+j2\pi f} = \frac{2a}{a^2+(2\pi f)^2}$$

$$|X(f)| = \frac{2a}{a^2+(2\pi f)^2}$$

$$\theta_X(f) = 0$$



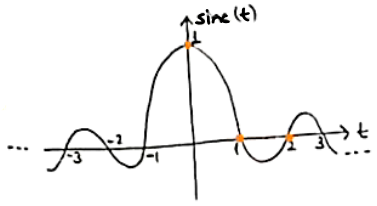
Ex: $x(t) = \begin{cases} 1, & |t| < \tau/2 \\ 0, & \text{dışında} \end{cases}$ dikdörtgen darbe işaretinin FT bulunuz.



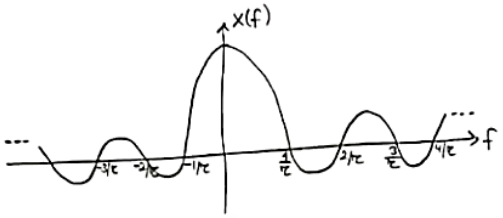
$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\,dt = \int_{-\tau/2}^{\tau/2} e^{-j2\pi ft}\,dt = \frac{e^{-j2\pi ft}}{-j2\pi f}\Bigg|_{-\tau/2}^{\tau/2}$$

$$= \frac{-1}{j2\pi f}\left[e^{-j2\pi f\tau/2} - e^{+j2\pi f\tau/2}\right] = \frac{1}{\pi f}\sin(\pi f\tau)$$

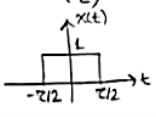
Reminder: $\operatorname{sinc}(t) = \begin{cases} \dfrac{\sin(\pi t)}{\pi t}, & t \neq 0 \\ 1, & t = 0 \end{cases}$



$$X(f) = \frac{1}{\pi f}\sin(\pi f\tau) = \tau\,\frac{\sin(\pi f\tau)}{\pi f\tau} = \tau\operatorname{sinc}(f\tau)$$



$$x(t) = \Pi\left(\frac{t}{\tau}\right) \xleftrightarrow{\mathcal{F}} X(f) = \tau\operatorname{sinc}(f\tau)$$

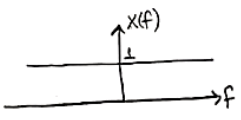


Ex: $x(t) = \delta(t)$ ise $X(f) = ?$

$$X(f) = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi ft}\,dt = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi f\cdot 0}\,dt = \int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

$$x(t) = \delta(t) \xleftrightarrow{\mathcal{F}} X(f) = 1$$





Ex: $X(f)=\delta(f-f_0)$ ise $x(t)=?$

$$x(t)=\int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df=\int_{-\infty}^{\infty}\delta(f-f_0)e^{j2\pi ft}df=\int_{-\infty}^{\infty}\delta(f-f_0)e^{j2\pi f_0 t}df=e^{j2\pi f_0 t}\underbrace{\int_{-\infty}^{\infty}\delta(f-f_0)df}_{=1}=e^{j2\pi f_0 t}$$

Ex: $X(f)=\delta(f)$ ise $x(t)=?$

$$x(t)=\int_{-\infty}^{\infty}X(f)e^{j2\pi ft}df=\int_{-\infty}^{\infty}\delta(f)e^{j2\pi ft}df=\int_{-\infty}^{\infty}\delta(f)e^{j2\pi 0t}df=\int_{-\infty}^{\infty}\delta(f)df=1$$

$$\delta(t)\xleftrightarrow{FF}1$$

$$1\xleftrightarrow{FF}\delta(f)$$

$$e^{j2\pi f_0 t}\xleftrightarrow{FF}\delta(f-f_0)$$

$$e^{-j2\pi f_0 t}\xleftrightarrow{FF}\delta(f+f_0)$$

Periyodik işaretlerin Fourier Dönüşümü

$x(t)=x(t+T),\ \forall t \qquad T=\frac{1}{f_0}=\frac{2\pi}{\omega_0}$

$$x(t)=\sum_{k=-\infty}^{\infty}x_k e^{jk\omega_0 t}=\sum_{k=-\infty}^{\infty}x_k e^{j2\pi kf_0 t}=\cdots+x_{-1}e^{-j2\pi f_0 t}+x_0+x_1e^{j2\pi f_0 t}+x_2e^{j2\pi 2f_0 t}+\cdots$$

($\omega_0=2\pi f_0$)

$$x_1e^{j2\pi f_0 t}\xleftrightarrow{FF}x_1\delta(f-f_0)$$

$$x_0\xleftrightarrow{FF}x_0\delta(f)$$

$$x_2e^{j2\pi 2f_0 t}\xleftrightarrow{FF}x_2\delta(f-2f_0)$$

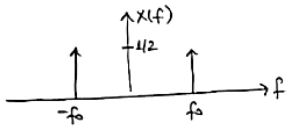
$$x(t)=\sum_{k=-\infty}^{\infty}x_k e^{j2\pi kf_0 t}\xleftrightarrow{FF}X(f)=\sum_{k=-\infty}^{\infty}x_k\delta(f-kf_0)$$

$$x_k=\frac{1}{T_0}\int_{T_0}x(t)e^{-jk\omega_0 t}dt$$

Ex: $x(t)=\cos 2\pi f_0 t$ ise $X(f)=?$

$$x(t)=\frac{1}{2}\left[e^{j2\pi f_0 t}+e^{-j2\pi f_0 t}\right]\longrightarrow x_1=x_{-1}=1/2$$

$$X(f)=\frac{1}{2}\delta(f-f_0)+\frac{1}{2}\delta(f+f_0)$$

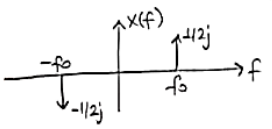


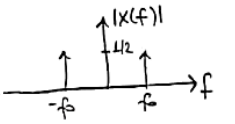
Ex: $x(t)=\sin 2\pi f_0 t$

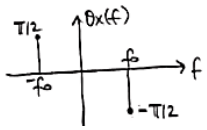
$$x(t)=\frac{1}{2j}\left[e^{j2\pi f_0 t}-e^{-j2\pi f_0 t}\right]\longrightarrow x_1=\frac{1}{2j}=-\frac{1}{2}j,\quad x_{-1}=\frac{-1}{2j}=\frac{1}{2}j$$

$$|x_1|=|x_{-1}|=1/2,\quad \theta_{x_1}=-\pi/2,\quad \theta_{x_{-1}}=\pi/2$$

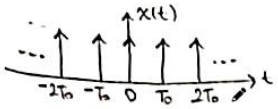
$$X(f)=\frac{1}{2j}\left[\delta(f-f_0)-\delta(f+f_0)\right]$$





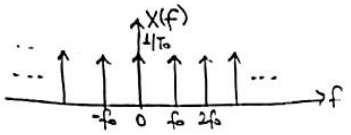


Ex: $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_0)$ dürtü katarı işaretinin FT bulunuz.



Reminder: $x_k = \frac{1}{T_0}\int_{T_0} x(t)e^{-jk\omega_0 t} = \frac{1}{T_0}$

$$X(f) = \sum_{k=-\infty}^{\infty} x_k \delta(f - kf_0) = \frac{1}{T_0}\sum_{k=-\infty}^{\infty} \delta(f - kf_0)$$



Sürekli Zamanlı Fourier Transformunun Özellikleri

$x(t) \xleftrightarrow{\mathcal{F}} X(f)$, $x(t)$: aperiyodik işaret

$X(f) = \mathcal{F}\{x(t)\}$

1) Lineerlik: $\alpha x_1(t) + \beta x_2(t) \xleftrightarrow{\mathcal{F}} \alpha X_1(f) + \beta X_2(f)$

2) Zamanda öteleme: $x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-j2\pi f t_0} X(f)$

$$\mathcal{F}\{x(t-t_0)\} = \int_{-\infty}^{\infty} x(t-t_0)e^{-j2\pi f t}dt \underset{t - t_0 = \tau}{=} \int_{-\infty}^{\infty} x(\tau)e^{-j2\pi f(\tau + t_0)}d\tau = e^{-j2\pi f t_0}\underbrace{\int_{-\infty}^{\infty} x(\tau)e^{-j2\pi f\tau}d\tau}_{X(f)}$$

Aperiyodik bir $x(t)$ işareti zamanda ötelendiğinde genlik spektrumu değişmez, faz spektrumu $-2\pi f t_0$ kadar ötelenir.

3) Frekansta öteleme: $x(t)e^{j2\pi f_0 t} \xleftrightarrow{\mathcal{F}} X(f - f_0)$

İspat: $\mathcal{F}\{x(t)e^{j2\pi f_0 t}\} = \int_{-\infty}^{\infty} x(t)e^{j2\pi f_0 t}e^{-j2\pi f t}dt = \int_{-\infty}^{\infty} x(t)e^{-j2\pi(f - f_0)t}dt = X(f - f_0)$

$x(t)$ işareti $e^{j2\pi f_0 t}$ ile çarpıldığında frekans spektrumu f_0 kadar ötelenir.

4) Modülasyon: $x(t)\cos 2\pi f_0 t \xleftrightarrow{\mathcal{F}} \frac{1}{2}[X(f - f_0) + X(f + f_0)]$

$$x(t)\sin 2\pi f_0 t \xleftrightarrow{\mathcal{F}} \frac{1}{2j}[X(f - f_0) - X(f + f_0)] = \frac{1}{2}\left[X(f - f_0)e^{-j\pi/2} + X(f + f_0)e^{j\pi/2}\right]$$

5) Zamanda ölçeklendirme: $x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|}X(f/a)$, $a \neq 0$

$\longrightarrow a > 1$ işaret frekansta (spektrum) genişler.

$0 < a < 1$ işaret frekansta daralır.

6) Duality: $x(t) \xleftrightarrow{\mathcal{F}} X(f)$

$X(t) \xleftrightarrow{\mathcal{F}} x(-f)$

İspat: $x(t) = \mathcal{F}^{-1}\{X(f)\} = \int_{-\infty}^{\infty} X(f)e^{j2\pi f t}df$

$$x(-t) = \int_{-\infty}^{\infty} X(f)e^{-j2\pi f t}df \underset{t \to f}{\Rightarrow} x(-f) = \int_{-\infty}^{\infty} X(t)e^{-j2\pi f t}dt$$

7) Türev alma: $\frac{dx(t)}{dt} \xleftrightarrow{\mathcal{F}} j2\pi f X(f)$

$$\frac{d^n x(t)}{dt^n} \xleftrightarrow{\mathcal{F}} (j2\pi f)^n X(f)$$

İspat: $\mathcal{F}\left\{\frac{dx(t)}{dt}\right\} = \mathcal{F}\left\{\frac{d}{dt}\underbrace{\left[\int_{-\infty}^{\infty} X(f)e^{j2\pi ft}dt\right]}_{x(t)=\mathcal{F}^{-1}\{X(f)\}}\right\} = \mathcal{F}\left\{j2\pi f \underbrace{\int_{-\infty}^{\infty} X(f)e^{j2\pi ft}dt}_{x(t)}\right\} = j2\pi f\,\mathcal{F}\{x(t)\} = j2\pi f X(f)$

8) Integral alma: $\int_{-\infty}^{t} x(\tau)d\tau \xleftrightarrow{\mathcal{F}} \frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\delta(f)$

9) Konvolüsyon: $x_1(t) * x_2(t) \xleftrightarrow{\mathcal{F}} X_1(f).X_2(f)$

10) Çarpma: $x_1(t).x_2(t) \xleftrightarrow{\mathcal{F}} X_1(f) * X_2(f)$

11) Eşlenik alma (conjugation): $x^*(t) \xleftrightarrow{\mathcal{F}} X^*(-f)$

⇒ $x(t)$ reel ise FT eşlenik simetriktir.

$x(t) = x^*(t) \xleftrightarrow{\mathcal{F}} X(f) = X^*(-f)$

⇒ $x(t)$ reel ve çift ise FT reel ve çifttir.

$x(t) = x^*(t) \xleftrightarrow{\mathcal{F}} X^*(f) = X(f)$

$x(t) = x(-t) \qquad X(f) = X(-f)$

⇒ $x(t)$ reel ve tek ise FT sanal ve tektir.

$x(t) = x^*(t) \xleftrightarrow{\mathcal{F}} X^*(f) = -X(f)$

$x(t) = -x(-t) \qquad X(-f) = -X(f)$

Parseval teoremi:

$$E_x = \int_{-\infty}^{\infty}|x(t)|^2dt = \int_{-\infty}^{\infty}|X(f)|^2df \quad \longleftarrow |X(f)|^2: \text{Enerji spektral yoğunluğu (energy spectral density)}$$

İspat: $\int_{-\infty}^{\infty}|x(t)|^2dt = \int_{-\infty}^{\infty}x(t)x^*(t)dt = \int_{-\infty}^{\infty}x(t)\left[\int_{-\infty}^{\infty}X^*(f)e^{-j2\pi ft}df\right]dt = \int_{-\infty}^{\infty}X^*(f)\underbrace{\left[\int_{-\infty}^{\infty}x(t)e^{-j2\pi ft}dt\right]}_{X(f)}df$

$$= \int_{-\infty}^{\infty}X^*(f)X(f)df = \int_{-\infty}^{\infty}|X(f)|^2df$$

Ex: $x(t) = A\tau\Pi(t/\tau) \xleftrightarrow{\mathcal{F}} X(f) = A\tau\,\text{sinc}(f\tau)$ ise $x(t) = \text{sinc}(t\tau)$ işaretinin FT=?

Reminder: $x(t) \longleftrightarrow X(f)$, $X(f) \longleftrightarrow x(-t)$ } Duality

$x(t) = A\tau\,\text{sinc}(t\tau) \xleftrightarrow{\mathcal{F}} X(f) = A\Pi(-f/\tau) = A\Pi(f/\tau)$ (↑ çift fonksiyon)

$x(t) = \text{sinc}(t\tau) \xleftrightarrow{\mathcal{F}} X(f) = \frac{1}{\tau}\Pi(f/\tau)$

Ex: $x(t) = \frac{2}{1+t^2}$ ise $X(w) = ?$

Reminder: $e^{-a|t|} \longleftrightarrow \frac{2a}{a^2+(w)^2}$

$x(t) = e^{-|t|} \longleftrightarrow X(w) = \frac{2}{1+(w)^2}$

$x(t) = \frac{2}{1+t^2} \longleftrightarrow X(w) = e^{-|-w|} = e^{-|w|}$

Ex: $\mathcal{F}\{u(t)\} = ?$

Reminder: $u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \xleftrightarrow{\mathcal{F}} \frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\delta(f)$$

$$\mathcal{F}\{u(t)\} = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$$

$x(t) = \delta(t)$

$X(f) = 1$

Ör: $x(t) = \frac{1}{1+t^2}$ Find the fourier transform.

$$e^{-\alpha|t|} \longrightarrow \frac{2\alpha}{\alpha^2+(2\pi f)^2} = \frac{2\alpha}{4\pi^2}\cdot\frac{1}{\frac{\alpha^2}{4\pi^2}+f^2}$$

$x(t) \longrightarrow X(f)$
$X(t) \longrightarrow x(-t)$ } Duality

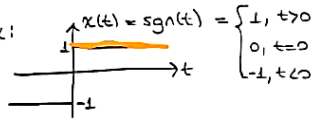
$$X(t) = \frac{1}{1+t^2} \xrightarrow{\mathcal{F}} x(-f) \qquad \frac{2\alpha}{4\pi^2}\,\frac{1}{\frac{\alpha^2}{4\pi^2}+t^2} \xrightarrow{\mathcal{F}} e^{-\alpha|-f|} = e^{-\alpha|f|}$$

$$\frac{1}{\frac{\alpha^2}{4\pi^2}+t^2} \longrightarrow \frac{4\pi^2}{2\alpha}\cdot e^{-\alpha|f|}$$

$\frac{\alpha^2}{4\pi^2} = 1 \quad \alpha^2 = 4\pi^2 \rightarrow \alpha = 2\pi$

$$\frac{1}{1+t^2} \xrightarrow{\mathcal{F}} \pi e^{-2\pi|f|}$$

Ex: $x(t) = sgn(t) = \begin{cases} 1, & t>0 \\ 0, & t=0 \\ -1, & t<0 \end{cases}$ ise $X(f) = ?$



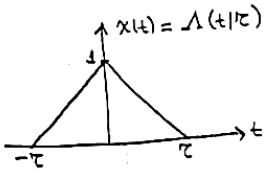
$x(t) = sgn(t) = u(t) - u(-t)$

Reminder: $x(at) \longleftrightarrow \frac{1}{|a|}X(f/a)$

$$\mathcal{F}\{sgn(t)\} = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f) - \frac{1}{|-1|}\left[\frac{1}{j2\pi(-f)} + \frac{1}{2}\delta(-f)\right] = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f) + \frac{1}{j2\pi f} - \frac{1}{2}\delta(f) = \frac{1}{j\pi f}$$

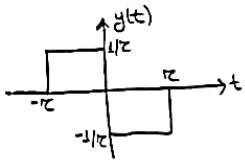
$\delta(f) = \delta(-f)$
Çift fonksiyon

Ex: $x(t) = \Lambda(t/\tau)$ üçgen işaretinin FT bulunuz.



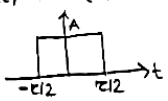
$$x(t) = \begin{cases} 1 + \frac{t}{\tau}, & -\tau < t < 0 \\ 1 - \frac{t}{\tau}, & 0 < t < \tau \end{cases}$$

$$y(t) = \frac{dx(t)}{dt} = \begin{cases} 1/\tau, & -\tau < t < 0 \\ -1/\tau, & 0 < t < \tau \end{cases}$$



Reminder: $\frac{dx(t)}{dt} \longleftrightarrow j2\pi f X(f) \Rightarrow Y(f) = j2\pi f X(f) \rightarrow X(f) = \frac{1}{j2\pi f} Y(f)$

Reminder: $z(t) = A\Pi(t/\tau) \xleftrightarrow{FF} Z(f) = A\tau \,\mathrm{sinc}(f\tau)$



$A = 1/\tau \rightarrow z(t) = \frac{1}{\tau}\Pi(t/\tau) \longleftrightarrow Z(f) = \mathrm{sinc}(f\tau)$

$y(t) = z(t+\tau/2) - z(t-\tau/2)$

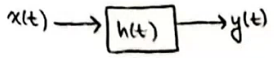
$Y(f) = Z(f)e^{j2\pi f\tau/2} - Z(f)e^{-j2\pi f\tau/2}$

$X(f) = \frac{1}{j2\pi f} Y(f) = \frac{1}{j2\pi f} Z(f)\left[e^{j\pi f\tau} - e^{-j\pi f\tau}\right] = \frac{Z(f)}{\pi f}\sin(\pi f\tau) = Z(f)\,\tau\,\mathrm{sinc}(f\tau) = \tau\,\mathrm{sinc}^2(f\tau)$

$x(t) \triangleq \Lambda(t/\tau) \xleftrightarrow{FF} X(f) \triangleq \tau\,\mathrm{sinc}^2(f\tau)$

10. Hafta

LTI Sistemlerin Frekans Domeni Analizi



$y(t) = x(t) * h(t) \xleftrightarrow{FF} Y(f) = X(f)\,H(f)$

- $h(t)$: dürtü cevabı (impulse response)
- $H(f)$: frekans cevabı (frequency response)

$H(f) = |H(f)|\,e^{j\theta_h(f)}$

- $|H(f)|$: genlik cevabı (amplitude response)
- $\theta_h(f)$: faz cevabı (phase response)

$|Y(f)|\,e^{j\theta_Y(f)} = |X(f)|\,e^{j\theta_X(f)}\,|H(f)|\,e^{j\theta_h(f)}$

- $|Y(f)| = |X(f)|\,|H(f)|$
- $\theta_Y(f) = \theta_X(f) + \theta_h(f)$

Ex: $x(t) = e^{-at}u(t)$, $a>0$ ise $y(t) = ?$

$h(t) = e^{-bt}u(t)$, $b>0$

$y(t) = x(t) * h(t) = \mathcal{F}^{-1}\{Y(f) = X(f)H(f)\}$

Reminder: $e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a+j2\pi f}$

$$Y(f) = \frac{1}{(a+j2\pi f)(b+j2\pi f)} = \frac{A}{a+j2\pi f} + \frac{B}{b+j2\pi f} \quad \leftarrow A = -B = \frac{1}{b-a}$$

$$Y(f) = \frac{1}{b-a}\left[\frac{1}{a+j2\pi f} - \frac{1}{b+j2\pi f}\right] \xleftrightarrow{\mathcal{F}^{-1}} y(t) = \frac{1}{b-a}\left[e^{-at}u(t) - e^{-bt}u(t)\right]$$

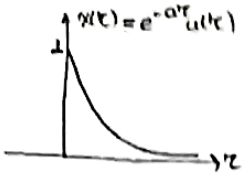
Ex: $x(t) = h(t) = e^{-at}u(t)$, $a>0$ ise $y(t) = ?$

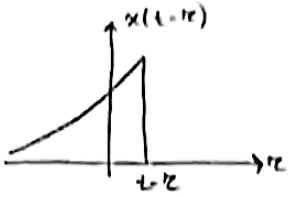
$$Y(f) = \left(\frac{1}{a+j2\pi f}\right)^2$$

Reminder: $e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a+j2\pi f}$

$x_1(t) * x_2(t) \xleftrightarrow{\mathcal{F}} X_1(f) \cdot X_2(f)$

$y(t) = \mathcal{F}^{-1}\{X(f)X(f)\} = x(t) * x(t) = e^{-at}u(t) * e^{-at}u(t)$





$$y(t) = x(t) * x(t) = \int_{-\infty}^{\infty} e^{-a\tau}u(\tau)\, e^{-a(t-\tau)}u(t-\tau)\,d\tau = e^{-at}\int_0^t 1\,d\tau = te^{-at}, \quad t>0$$

$$te^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{(a+j2\pi f)^2}$$

Ex: $x(t) = e^{-2t}u(t)$ ise sistemin frekans cevabını ve dürtü cevabını bulunuz.

$y(t) = e^{-t}u(t)$

$$Y(f) = X(f)H(f) \longrightarrow H(f) = \frac{Y(f)}{X(f)} \xleftrightarrow{\mathcal{F}^{-1}} h(t)$$

$$H(f) = \frac{\frac{1}{1+j2\pi f}}{\frac{1}{2+j2\pi f}} = \frac{2+j2\pi f}{1+j2\pi f} = 1 + \frac{1}{1+j2\pi f} \xleftrightarrow{\mathcal{F}^{-1}} h(t) = \delta(t) + e^{-t}u(t)$$

Reminder: $1 \longleftrightarrow \delta(f)$

$\delta(t) \longleftrightarrow 1$

Ex: $\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 8y(t) = 2x(t)$ ile tanımlanan LTI sistemin dürtü cevabını bulunuz.

$$(j2\pi f)^2 Y(f) + 6j2\pi f Y(f) + 8Y(f) = 2X(f)$$

$$H(f) = \frac{Y(f)}{X(f)} = \frac{2}{-(2\pi f)^2 + 6j(2\pi f) + 8} \overset{w=2\pi f}{=} \frac{2}{-w^2 + 6jw + 8} = \frac{2}{(jw+4)(jw+2)} = \frac{A}{jw+4} + \frac{B}{jw+2}$$

$$\begin{matrix} jw & 4 \\ jw & 2 \end{matrix}$$

$$A = B = -1$$

$$H(f) = \frac{1}{jw+2} - \frac{1}{jw+4} \xleftrightarrow{\mathcal{F}^{-1}} h(t) = \left(e^{-2t} - e^{-4t}\right)u(t)$$